

Improvement of high-cycle fatigue life in a 6061 Al alloy produced by equal channel angular pressing

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Abstract

In the present study, mechanical properties and fatigue behavior of solid solution treated 6061 Al alloy fabricated by equal channel angular pressing (ECAP) process were evaluated. Yield stress and ultimate tensile stress significantly increased after ECAP. A remarkably large enhancement in fatigue life, by a factor of about 10, compared to a T6 treated commercial 6061 Al alloy was recognized to occur in low and high cycle regimes after a single pass. Further deformation by ECAP, however, virtually eliminated this improvement especially in the high cycle regime. Fine-grained microstructure with low-grain boundary misorientation angles was proposed to yield the best result in fatigue performance in a 6061 Al alloy. The current result suggests that one may pay attention to a single passed material rather than multi-passed material if improvement in fatigue life is a primary concern for successful engineering applications. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Fatigue life; Grain boundary angle; Equal channel angular pressing

1. Introduction

Equal channel angular pressing (ECAP) is an innovative method of introducing intensive plastic strain into materials through repetitive pressing [1–4]. The ECAP processed materials typically have very small grain sizes less than 1 μm and exhibit superior mechanical properties such as yield stress, ultimate tensile strength [5,6]. While there has been considerable emphasis on these mechanical properties, there have been relatively few numbers of studies on fatigue behavior [7–10]. Recently, fatigue properties of an ECAP processed 5056 Al alloy (4 pass and 8 pass) were studied under a fully reversed constant stress amplitude condition [11,12] and it was reported that ECAP technique did not reveal a distinct advantage over standard processing in the high cycle regime [11,12]. The purpose of the current study is to investigate a fatigue behavior of a 6061 Al alloy processed by ECAP, which is one of the aluminum

alloys being massively used in aerospace and automobile industries, and explore a possibility of improving fatigue life of the ECAP processed 6061 Al alloy in the high cycle regime.

2. Experimental procedures

For the sample preparation, a 6061 Al billet was extruded to a diameter of 18 mm at 450 °C and cut to the length of 100 mm. The extruded bar was solid-solutionized at 530 °C and then quenched into water. This heat-treated material was held at 125 °C for 20 min and then pressed through the die (channel intersection angle of 90°) preheated to 125 °C. Repetitive pressings of the same sample were performed up to 4 pass. All pressings were conducted by rotating each sample about the longitudinal axis by 90° in the same direction between consecutive passes (designated as route B_c [1]).

Tensile tests were carried out to characterize the strength and ductility of the ECAP processed 6061 Al alloys. Tensile test was operated at a constant cross-head speed with an initial strain rate of $5 \times 10^{-4} \text{ s}^{-1}$.

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Fatigue life experiments were carried out on the specimens (diameter = 8 mm) with dimension ratios based on the ASTM standard E 606-92 under the complete unloading condition $R = 0$ (i.e. positive mean stress loading condition), where R is the ratio of the minimum stress to maximum stress for a cycle stress. This loading condition usually provides more severe fatigue environment than the fully reversed loading condition. In real application related to energy generation and transition, such as turbines, pressure vessels, and pipelines, fatigue failure usually results from this kind of loading condition [13].

Crack-propagation behavior was characterized using single edge notched (SEN, $W = 18$ mm, $a_o = 5$ mm, $t = 2$ mm) samples at the stress ratio $R = 0.1$. The stress intensity factor for the SEN sample was calculated using the following equation:

$$K = \left(\frac{P}{Wt} \right) (\pi a)^{0.5} \times \left\{ 1.12 - 0.231 \left(\frac{a}{W} \right) + 10.55 \left(\frac{a}{W} \right)^2 - 21.72 \left(\frac{a}{W} \right)^3 + 30.39 \left(\frac{a}{W} \right)^4 \right\} \quad (1)$$

where W , t and a are the width and thickness of specimen and crack length, respectively. A travelling microscope with a magnification of 100 was used for measuring the crack length. The crack growth rate, da/dN , where N is the fatigue number of cycle was computed using the incremental polynomial method (ASTM E647). For the fatigue life and crack propagation tests, a sinusoidal waveform load control mode was employed with frequency of 10 Hz.

Microstructures of the ECAP processed samples were examined using transmission electron microscopy (TEM). Samples for TEM were cut from the y plane that is parallel to the side faces at the point of exit from the die and selected area electron diffraction (SAED) patterns were taken from areas having a diameter of ~ 10 μm .

3. Results

3.1. Microstructure of the ECAP processed 6061 Al alloy

TEM micrographs of the billets after 1 and 4 pass are shown in Fig. 1(a) and (b), respectively, together with SAED patterns. The grain size of the undeformed material determined by optics was as coarse as ~ 40 – 80 μm . After 1 pass, elongated dislocation cells or subgrains have formed parallel to the shear direction. Further pressing (4 pass) has fragmented the elongated substructure to equiaxed grains (~ 0.3 – 0.4 μm) (Fig.

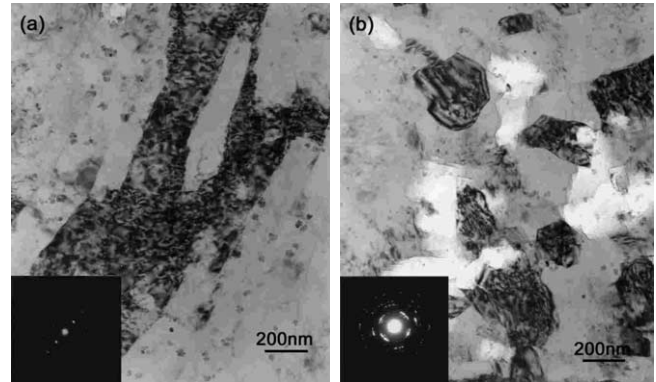


Fig. 1. TEM micrographs of the 6061 Al alloys after ECAP process (a) 1 pass (b) 4 pass.

1(b)). The SAED pattern shown in Fig. 1(b) exhibits the diffracted beams scattered more uniformly around rings than that of the 1 pass material in Fig. 1(a). This observation indicates that the number of subgrains and the misorientation angle between subgrain boundaries have increased with pass number by forming new boundaries perpendicular to the shear direction and depositing new dislocations into existing subgrain boundaries.

3.2. Stress–strain curves

The stress–strain curves obtained from the tensile tests on the 0, 1 and 4 pass 6061Al alloy are presented in Fig. 2. It is obvious that ECAP process resulted in a significant increase in both yield stress and ultimate tensile strength. The yield stress has increased about 112% only after 1 pass. After 4 pass, the increment reaches near 156%. This large increase in strength is due to the considerable grain refinement made through severe deformation by ECAP. Tensile ductility, on the other hand, has decreased by about half after ECAP. This reduction in ductility after ECAP process is

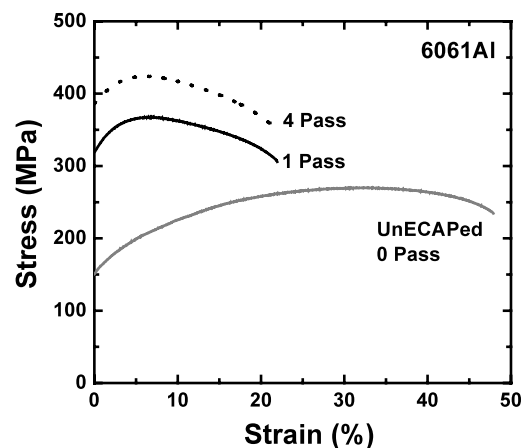


Fig. 2. Stress–strain curves for the unECAPed and ECAPed 6061Al alloy.

commonly observed in many ECAP processed alloys and considered to be due to the decrease in hardening exponent with increase in accumulated strain.

3.3. Fatigue life

Fatigue life of the alloys in the current investigation was assessed through the stress amplitude (σ_a) versus number of cycles ($S-N$ curve). Fig. 3 is the $S-N$ curves for the 6061 Al alloy at 0, 1 and 4 pass. The fatigue-life data of a T6 treated commercial 6061 Al alloy tested under the same loading condition ($R = 0$) [14] are also plotted for comparison, which turn out to be similar to those of the 0 pass material. After a single pass, there took place a significant increase in fatigue life, by a factor of more than ten, in both low and high cycle regimes. When further pressed (4 pass), however, such fatigue improvement in the high cycle regime became largely reduced. This feature of little improvement in the high-cycle fatigue performance in the 4 pass material is similar to that reported on the 4 pass 5056 Al alloy [11].

3.4. Crack propagation

Crack propagation behavior of the ECAP processed 6061 Al alloy was assessed through crack growth rate versus ΔK , where K is the stress intensity factor. The crack growth rate, da/dN , where a is the crack size, is plotted for the 0, 1, and 4 pass materials as a function of ΔK in double log scales (Fig. 4). The following can be inferred from the plot. First, after a single pass, little change in crack-growth-behavior is resulted, which is true over the entire range of ΔK . It slightly, however, decreased after 4 pass at small ΔK values ($\leq 7 \text{ MPa m}^{1/2}$). Second, extrapolation of the data to smaller ΔK values ($\sim 4\text{--}5 \text{ MPa m}^{1/2}$) indicates that there is a trend

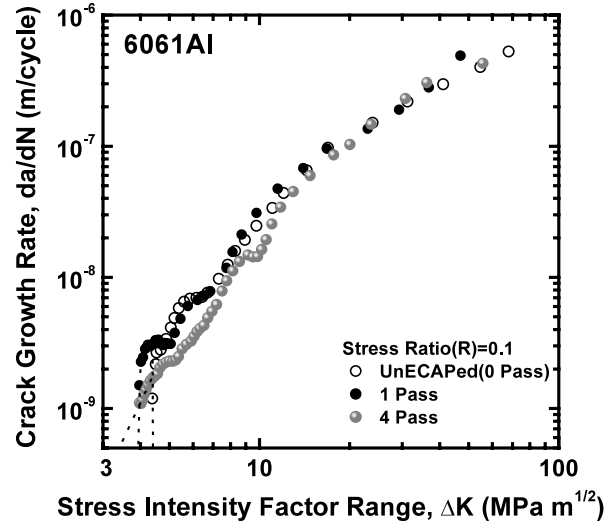


Fig. 4. Fatigue crack growth rate as a function of ΔK .

that ECAPed materials exhibit a lower fatigue threshold than the unECAPed material. A similar threshold trend was also observed in the ECAP processed 5056 Al alloy [11]. These observations are in line with the common observation that the material with coarse grains shows a higher fatigue threshold [15–17].

4. Discussion

It is known that grain refinement increases the resistance of fatigue crack initiation by reducing the number of dislocations in slip bands that can lead to a decrease of damage depth of surface intrusions or extrusions, or by increasing the yield stress that prevents macroscopic plastic deformation at the beginning of load controlled cycling [15–17]. Resistance to crack growth, on the other hand, is known to decrease with decrease in grain size since less tortuous crack path is induced during fatigue crack growth [16], which causes decrease in driving force for crack propagation. Therefore, it is anticipated in general that fatigue life increases by grain refinement when the effect of increased resistance to crack nucleation is dominant over the effect of decreased resistance to crack propagation.

Vinogradov et al. [11], who compared the fatigue properties of the 0 and 4 pass processed 5056 Al alloys, attributed to the increased fatigue life of the 4 pass 5056 Al alloy in the low cycle regime, where high stresses were applied, to the increased resistance to crack nucleation by its increased yield strength. Observation of crack growth rate that changed little at large ΔK values ($\geq 8 \text{ MPa m}^{1/2}$) after 4 pass by ECAP supported this view. Little improvement in fatigue life observed in the high cycle regime where low stresses were applied, on the other hand, was ascribed to elimination of an advantage of the increased resistance against crack nucleation by

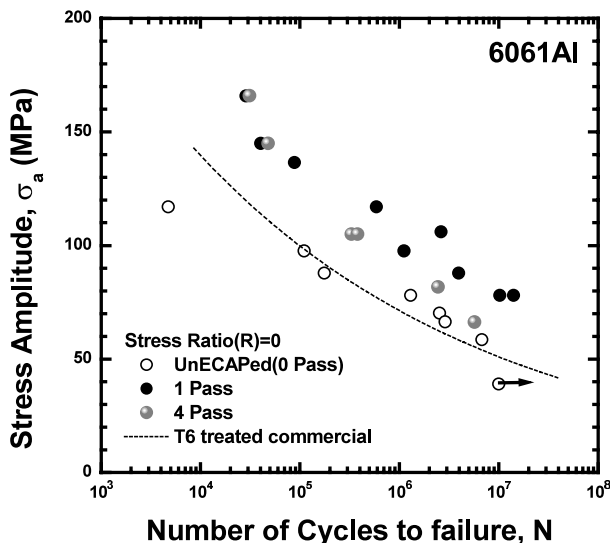


Fig. 3. $S-N$ curves for the unECAPed and ECAPed 6061Al alloy.

the increased crack growth rates at small ΔK values ($\leq 8 \text{ MPa m}^{1/2}$) after ECAP [11]. When the fatigue-life behavior of the 1 pass 6061 Al alloy is similarly evaluated based on its crack growth behavior, it is natural to expect the fatigue-life enhancement in low as well as high cycle regimes. This is because crack growth rate of the 1 pass 6061 Al material is similar to that of the 0 pass material over the entire ΔK range and thus the effect of increased resistance to crack nucleation by ECAP is sure to dominate the fatigue life. Then, why is the fatigue life of the 4 pass 6061 Al alloy significantly shorter than that of the 1 pass alloy in the high cycle regime regardless of its lower crack growth rate compared to the 1 pass material at small ΔK values? This result may indicate, though the reason for the difference in crack growth behavior at small ΔK values between the 5056 and 6061 Al alloys at 4 pass is not clear at present, that the crack growth rate does not significantly affect the fatigue life behavior. This speculation agrees with the well-know fact that crack propagation phase of the high cycle fatigue life is relatively short compared to the nucleation phase such that the nucleation phase usually determines the fatigue life of a material [10]. We will, therefore, consider two factors possibly lowering resistance of crack nucleation in the 4 pass 6061 Al alloy in the high cycle regime: (1) cyclic softening and (2) grain-boundary misorientation angle.

If cyclic softening, which is the process whereby a material loses strength during cyclic loading, occurs more severely in the 4 pass alloy than in the 1 pass alloy, one may explain the decreased resistance to crack nucleation at 4 pass. To check this possibility, Vickers hardness was measured on the 0, 1 and 4 pass materials prior to and after cyclic to failure after polishing the cross-section of samples. The result is summarized in Fig. 5. Significant H_v scattering occurs after cyclic loading, which can be attributed to the inhomogeneity of plastic deformation during cyclic loading [12]. As seen, the H_v value, which is averaged over 20 measurements, varies with fatigue cycle number. For the 0 pass material, cyclic hardening occurs in the whole cycle range. For the 1 pass material, on the other hand, a small degree of cyclic softening, indicating that partial recovery occurs during cyclic loading, occurs except at the lowest fatigue cycle in the current investigation where small hardening is observed. For the 4 pass material, the average hardness is almost insensitive to cycle number. It is interesting to note that the softening and hardening behavior shown by the current alloys resemble their stress–strain relations obtained from tensile testing (Fig. 2). This may be resulted since the current cyclic loading was conducted in a tensile stress mode. It is now obvious from Fig. 5, where the hardness of the 4 pass 6061 Al alloy is still higher than the 1 pass 6061 Al alloy even in the high cycle regime, that the shorter fatigue life of the 4 pass 6061 Al alloy compared

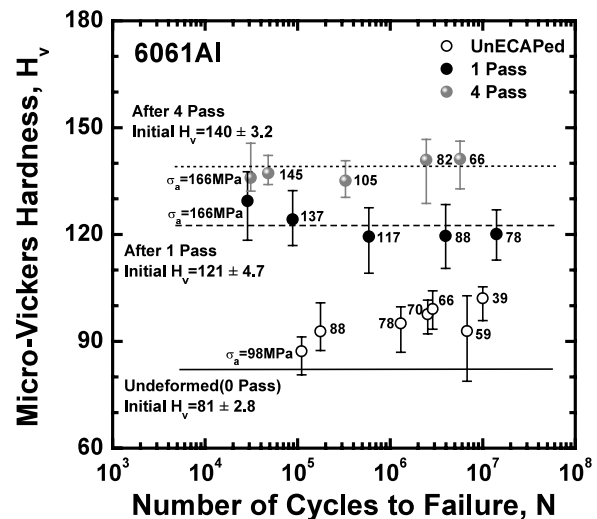


Fig. 5. Vickers hardness of the unECAPed and ECAPed 6061 Al alloys after fatigue tests.

to the 1 pass alloy in high cycle regime can not be ascribed to the effect of cyclic softening.

Zhang and Wang [18] studied crack-nucleation possibility along high and low-angle grain boundaries in a copper bicrystal with embedded grains and found that fatigue cracking preferentially took place along high-angle boundaries. As it is difficult for persistent slip bands to pass through the high angle grain boundaries, intergranular cracking by piling-up of dislocations at grain boundaries is more likely to occur at the higher-angle grain boundaries than at the lower-angle grain boundaries where dislocations can be transported into the adjacent grain by the shear bands [18]. If one refers to the observations by Zhang and Wang [18] and the fact that higher angle grain boundary has higher energy such that it is more vulnerable to intergranular cracking, it can be deduced that the difference in fatigue-life behavior between the 1 and 4 pass materials is a consequence of difference in grain-boundary misorientation angles. Note that the 4 pass material has higher angle boundaries than the 1 pass material (see the SAED patterns in Fig. 1(a) and (b)). Fig. 6 shows the SEM images of surfaces of 1 and 4 pass samples near the sites where fracture took place after cyclic failure at $\sigma_a = 105 \text{ MPa}$. Activity of persistent slip band can be observed in both samples. The bands in the samples are on a plane oriented approximately 45° from the loading axis that is the plane of the maximum resolved shear stress. Similar features of shear bands have been also observed in the ECAP processed Cu and 5056 Al alloy [9,11]. The observation on the surface of the ECAP processed 6061 Al alloys shows that multi micro-cracks that are parallel to the fractured surface are present in the 4 pass material, while these are absent in the 1 pass material. This observation indicates that micro-cracking more readily takes place in the 4 pass material than in the 1

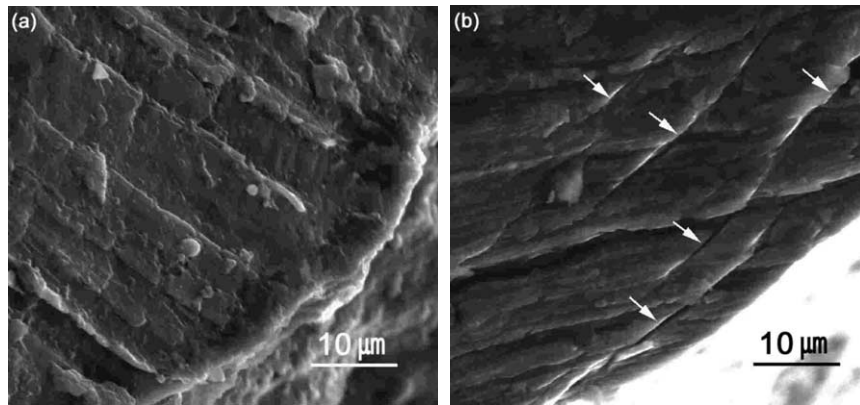


Fig. 6. Surface of the samples of the ECAPed 6061Al alloy near the place where fracture occurred after fatigue tests at $\sigma_a = 105\text{MPa}$ (a) 1 pass (b) 4 pass. Arrows indicates the presence of micro-cracks.

pass material and thus resistance to crack nucleation is possibly lower in the former material.

This effect of grain boundary angle on fatigue life, however, seems to be relatively insignificant in the low cycle regime where the fatigue lives of the 1 and 4 pass materials are revealed to be similar. In this regime, no micro-cracks were observed even in the 4 pass material. This observation implies that fatigue failure in the 1 and 4 pass materials proceeds without forming micro-cracks. In this case, neck formation was observed as in a typical tensile test where failure occurs by micro-void coalescence.

5. Conclusions

Mechanical properties and fatigue behavior of solid-solution treated 6061 Al alloy fabricated by ECAP process were assessed in tensile and cyclic experiments in terms of yield stress, ultimate tensile strength, tensile ductility, fatigue limit and crack propagation rate. It was found that:

(1) Yield stress and ultimate tensile stress of the 6061 Al alloy considerably increased after ECAP process, while tensile ductility decreased.

(2) A large enhancement in fatigue life by a factor of about 10 compared to a T6 treated commercial 6061 Al alloy was achieved after a single pass. This result is remarkable since the improvement was obtained in both low and high cycle regime. When the alloy was further pressed (4 pass), however, this advantage disappeared especially in the high cycle regime.

(3) The difference in fatigue life between the single passed and multi-passed alloy was attributed to the difference in grain boundary angle believed to affect the resistance of micro-crack nucleation during cyclic loading.

(4) The SEM observations on surfaces of the samples after cyclic failures show that micro-cracks are present in the 4 pass material, while they are not in the 1 pass

material. This can be a good evidence that the 4 pass material is more susceptible to micro-cracking than the 1 pass material such that the resistance to crack nucleation of the former alloy is lower than that of the latter alloy.

(5) The current finding suggests that a single passed material is better than the multi-passed material in fatigue performance. Though the strength of the 1 pass material is lower than that of the 4 pass material, it should be noted that its strength is still much higher than the undeformed material by a factor of more than 2. Therefore, when a balance between high strength and fatigue performance is a primary interest, a single passed material can be a good choice.

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